STATS 200A Lecture 12

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4 November 2026

Definition 12.1 (Bayes' Rule)

$$\mathbb{P}(x \mid y) = \frac{\mathbb{P}(y \mid x)\mathbb{P}_X(x)}{\mathbb{P}_Y(y)}, \qquad \mathbb{P}_Y(y) = \int \mathbb{P}(y \mid x)\mathbb{P}_X(x)\,dx.$$

$$\underbrace{\mathbb{P}(x \mid y)}_{\text{posterior for } x} \propto \underbrace{\mathbb{P}(y \mid x)}_{\text{likelihood}} \underbrace{\mathbb{P}_X(x)}_{\text{prior for } x}$$

$$\underbrace{\mathbb{P}(Y \in A, X \in B)}_{\mathbb{P}(X \in \cdot \mid Y = y)} = \mathbb{E}[\mathbf{1}_A(Y)\mathbf{1}_B(x)] = \mathbb{E}[\mathbf{1}_A(Y)K(Y,B)] = \int_A K(y,B)\nu(dy)$$

Remark 12.2 (The marginal and conditional aw determine the joint law)

Denote the measure with these rectangle probabilities by $\nu \otimes K$. Here it is the existing joint law: $\boxed{\mathbb{P}_{(Y,X)} = \nu \otimes K}$.

$$\mathbb{P}_{(Y,X)}(dy, dx) = K(y, dx)\nu(dy)$$

Sampling: draw $Y \sim \nu$, then $X \sim K(Y, \cdot)$. Coordinates are ordered (Y, X)

Example 12.3 (Gaussian posterior)

Prior: $X \sim N(m, s^2)$. Observation: $Y \mid X = x \sim N(x, \sigma^2); s, \sigma > 0$.
Equivalently, $Y = X + \sigma Z, Z \sim N(0,1)$ independent of X. Thus $Y \sim N(m, s^2 + \sigma^2)$.

$$X \mid Y = y \sim N(D^\star(y), v), \qquad D^\star(y) = \frac{s^2 y + \sigma^2 m}{s^2 + \sigma^2}, \quad v = \frac{s^2\sigma^2}{\sigma^2 + s^2}$$

$$X \sim N(m, s^2), Z \sim N(0,1), Y = X + \sigma Z$$

$$\begin{pmatrix} Z \\ X \\ Y \end{pmatrix} = N\left(\begin{pmatrix} 0 \\ m \\ m \end{pmatrix}, \begin{pmatrix} 1 & 0 & \sigma \\ 0 & s^2 & s^2 \\ \sigma & s^2 & \sigma^2 + s^2 \end{pmatrix}\right)$$

Remark 12.4 (Bayes: reweight the prior by the likelihood, then normalize)

$$\mathbb{P}_Y(y) = \int \mathbb{P}(y \mid X)\, \mu(dx), \qquad K(y, dx) = \frac{\mathbb{P}(y \mid x)}{\mathbb{P}_Y(y)}\, \mu(dx),$$

For $0 < \mathbb{P}_Y(y) < \infty$; the posterior has a density with respect to μ.